

Uniqueness of finite degree-one Boolean limit objects

A substantial partial answer to arXiv:1308.4108

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Status. Substantial partial result, likely valid. For Boolean-valued limit objects depending on finitely many degree-one coordinates, all affine-system densities determine a canonical finite aperiodic quotient up to affine isomorphism. The general fractional, higher-degree, and infinite-coordinate question remains open here.

1 Source question

The source is H. Hatami, P. Hatami, and J. Hirst, *Limits of Boolean Functions on $\mathbb{F}_p^n$* , arXiv:1308.4108 [1]. In the concluding remarks on source PDF page 12, it asks:

Suppose that two limit objects Γ_1, Γ_2 satisfy $t_{\mathcal{L}}(\Gamma_1) = t_{\mathcal{L}}(\Gamma_2)$ for every affine system of linear forms $\mathcal{L}$. Then how do Γ_1 and Γ_2 relate to each other?

6 Concluding remarks

We conclude with an open question regarding the uniqueness of the limit object. Suppose that two limit objects Γ_1, Γ_2 satisfy $t_{\mathcal{L}}(\Gamma_1) = t_{\mathcal{L}}(\Gamma_2)$ for every affine system of linear forms $\mathcal{L}$. Then how does Γ_1 and Γ_2 relate to each other?

The answer to the analogous question in the context of the limits of dense graphs is known. There the limit objects are represented as the so called graphons, which are symmetric measurable functions $W : [0, 1]^2 \rightarrow [0, 1]$. It is shown in [BCL10] that two graphons W_1, W_2 satisfy $t_H(W_1) = t_H(W_2)$ for every graph H if and only if there exists a third graphon W and two measure preserving maps $\sigma_1, \sigma_2 : [0, 1] \rightarrow [0, 1]$ such that for $i = 1, 2$, we have $W_i(x, y) = W(\sigma_i(x), \sigma_i(y))$ almost everywhere.

The source compares the desired relation with the common-factor characterization of graphons. The theorem below gives exactly such a common finite factor in a natural deterministic cylinder subclass.

2 Definitions and result

Fix a prime p . The source domain is

$$\mathbb{G}_{\infty} = \prod_{j \geq 1} \prod_{0 \leq k \leq \lfloor (j-1)/(p-1) \rfloor} \mathbb{U}_{k+1}^{\mathbb{N}}.$$

A *finite degree-one Boolean cylinder* is a measurable $\Gamma : \mathbb{G}_{\infty} \rightarrow \{0, 1\}$ of the form

$$\Gamma(x) = f(\pi(x)), \quad f : V \rightarrow \{0, 1\}, \quad V = \mathbb{F}_p^r, \quad (1)$$

where π selects finitely many coordinates of degree 1 and depth 0 (identifying $\mathbb{U}_1$ with $\mathbb{F}_p$). Redundant selected coordinates are allowed.

For a function $f : V \rightarrow \{0, 1\}$ define its translation-period subgroup

$$H_f = \{h \in V : f(x + h) = f(x) \text{ for every } x \in V\}.$$

It descends to a unique function $\bar{f} : W_f := V/H_f \rightarrow \{0, 1\}$. This descended function is *aperiodic*, meaning $H_{\bar{f}} = \{0\}$. We call $(W_f, \bar{f})$ the *minimal affine core* of the cylinder.

Theorem 1 (Finite degree-one uniqueness). *Let Γ_1, Γ_2 be finite degree-one Boolean cylinders, with minimal affine cores $(W_1, \bar{f}_1)$ and $(W_2, \bar{f}_2)$. The following are equivalent.*

1. *For every affine system of linear forms $\mathcal{L}$,*

$$t_{\mathcal{L}}(\Gamma_1) = t_{\mathcal{L}}(\Gamma_2).$$

2. *There is an affine bijection $T : W_1 \rightarrow W_2$ such that*

$$\bar{f}_1 = \bar{f}_2 \circ T.$$

3. *The two cylinders factor through a common finite affine Boolean object: there are a finite affine space W , a Boolean function $h : W \rightarrow \{0, 1\}$, and surjective affine maps q_i from their finite coordinate spaces to W such that $f_i = h \circ q_i$ for $i = 1, 2$.*

Thus the affine densities determine both the minimal core dimension and its affine-isomorphism class.

3 Proof intuition

Use as a test system all points of $\mathbb{F}_p^m$. On degree-one coordinates, a consistent sample of this system is exactly a uniformly random affine map $A : \mathbb{F}_p^m \rightarrow W$. Since the cylinder is Boolean-valued, the source's auxiliary Bernoulli sampling is deterministic, so the induced random object is the entire restriction array $\bar{f} \circ A$.

The translation periods of this array expose the rank of A . If $\bar{f}$ is aperiodic and A is onto, the period subgroup of $\bar{f} \circ A$ is exactly the kernel of the linear part of A . Consequently the smallest quotient dimension is detected by the first dimension in which an aperiodic restriction appears with positive probability. At that dimension, the aperiodic restrictions are precisely the affine orbit of $\bar{f}$. Equal densities give equal restriction laws, so they force equal dimensions and the same affine orbit.

4 Proof

For $m \geq 0$, let

$$\mathcal{A}_m = \{L_a : a \in \mathbb{F}_p^m\}, \quad L_a(x_0, x_1, \dots, x_m) = x_0 + \sum_{j=1}^m a_j x_j. \quad (2)$$

This is the affine system consisting of every point of the affine m -cube.

Lemma 1 (Restriction law). *Let Γ have minimal core $\bar{f} : W \rightarrow \{0, 1\}$. The random Boolean array on $\mathbb{F}_p^m$ induced by Γ and $\mathcal{A}_m$ has the same law as*

$$(\bar{f}(Aa))_{a \in \mathbb{F}_p^m}, \quad (3)$$

where $A : \mathbb{F}_p^m \rightarrow W$ is a uniformly random affine map.

Moreover, if two limit objects have equal $t_{\mathcal{L}}$ for every affine system $\mathcal{L}$, then their arrays (3) have the same law for every m .

Proof. For each selected degree-one coordinate, consistency with $\mathcal{A}_m$ means precisely that its values are

$$a \mapsto c_0 + \sum_{j=1}^m a_j c_j$$

for uniformly random independent coefficients $c_0, \dots, c_m \in \mathbb{F}_p$. Combining the finitely many coordinates gives a uniformly random affine map into the original coordinate space. Passing through the quotient by H_f still gives a uniformly random affine map into W , because a surjective linear quotient sends uniform independent coefficients to uniform independent coefficients. Since Γ takes only the values zero and one, the independent Bernoulli variables in the source definition equal their parameters almost surely. This proves (3).

For the final assertion, let $X = (X_a)_{a \in \mathbb{F}_p^m}$ be an induced Boolean array. For each $\omega \in \{0, 1\}^{\mathbb{F}_p^m}$, inclusion–exclusion gives

$$\begin{aligned} \Pr[X = \omega] &= \mathbb{E} \left[\prod_{\omega(a)=1} X_a \prod_{\omega(a)=0} (1 - X_a) \right] \\ &= \sum_{T \subseteq \{a : \omega(a)=0\}} (-1)^{|T|} t_{\{L_a : \omega(a)=1 \text{ or } a \in T\}}(\Gamma). \end{aligned}$$

Every system on the right is affine. Equality of all source densities therefore gives equality of every atom probability of the finite array law. $\square$

For a function $g : \mathbb{F}_p^m \rightarrow \{0, 1\}$ write

$$\text{Per}(g) = \{v \in \mathbb{F}_p^m : g(x + v) = g(x) \text{ for every } x \in \mathbb{F}_p^m\}.$$

Lemma 2 (Periods detect full rank). *Let $h : W \rightarrow \{0, 1\}$ be aperiodic, and let $A(x) = b + Lx : \mathbb{F}_p^m \rightarrow W$ be affine. Then*

$$\ker L \subseteq \text{Per}(h \circ A).$$

If L is surjective, equality holds. In particular, when $m = \dim W$, the restriction $h \circ A$ is aperiodic if and only if L is invertible.

Proof. The first inclusion is immediate. If L is surjective and $v \in \text{Per}(h \circ A)$, then

$$h(y + Lv) = h(y) \quad \text{for every } y \in W,$$

because $b + Lx$ ranges over all of W . Thus $Lv \in H_h = \{0\}$ and $v \in \ker L$. The last assertion follows because a square linear map is surjective exactly when it is invertible. $\square$

Proof of Theorem 1. Assume (1), and set $r_i = \dim W_i$. Suppose, toward a contradiction, that $r_1 < r_2$, and apply Lemma 1 with $m = r_2$. Every linear map $\mathbb{F}_p^{r_2} \rightarrow W_1$ has a nonzero kernel, so every restriction array from $\bar{f}_1$ has a nonzero period. For $\bar{f}_2$, the linear part is invertible with positive probability

$$\frac{|\mathrm{GL}(r_2, p)|}{p^{r_2^2}} > 0,$$

and Lemma 2 says that the corresponding array is aperiodic. The two finite array laws cannot be equal, contradicting Lemma 1. By symmetry $r_1 = r_2 =: r$.

Use $m = r$. For each core, Lemma 2 says that the aperiodic arrays are exactly those obtained from affine bijections $A : \mathbb{F}_p^r \rightarrow W_i$. Conditional on this positive-probability event, the support of the array law is therefore the affine orbit

$$\{\bar{f}_i \circ A : A \in \mathrm{Aff}(\mathbb{F}_p^r, W_i) \text{ is bijective}\}.$$

The conditional distribution is uniform on that orbit: the affine group acts transitively, and every orbit element has the same number of representing bijections (the stabilizer size). Equality of the full array laws forces the two orbit supports to coincide after identifying W_i with $\mathbb{F}_p^r$. Hence there is an affine bijection $T : W_1 \rightarrow W_2$ with $\bar{f}_1 = \bar{f}_2 \circ T$. This proves (2).

Statement (2) is equivalent to (3) by taking the common space to be either minimal core and composing one quotient map with T or T^{-1} .

Finally, (2) implies (1): composition with a fixed affine bijection sends a uniform random affine map to a uniform random affine map. Lemma 1 therefore gives identical restriction laws for every affine system, and in particular identical densities. $\square$

The constant-core case $r = 0$ is included: the only cores are the constant zero and constant one functions, and their one-point densities distinguish them.

5 Upgrade attempts and limitations

The proof was pushed in two directions beyond the promoted theorem.

Fractional finite cylinders. If Γ takes values in $[0, 1]$, a random affine map produces a mixture of independent product-Bernoulli arrays with parameter vector $(\Gamma(Aa))_a$, not that parameter vector itself. The event that the parameter array has trivial period is no longer directly observable. The full-rank extraction above becomes a latent-mixture identifiability problem; no unconditional separation argument was obtained.

Arbitrary Boolean measurable objects. Finite cylinders approximate such objects in measure, but the affine identifications furnished at successive finite scales need not be coherent. A full argument would require a coupling or inverse-limit theorem aligning the finite quotient cores.

Candela, González-Sánchez, and Szegedy give a deep representation of $\mathbb{F}_2^\omega$ -affine-exchangeable laws and a universal 2-adic limit domain in arXiv:2203.08915 [2]. They explicitly cite the source and clarify the law-level landscape, but the bounded search did not locate a theorem turning two particular source parameter functions with the same law into the direct graphon-style common-factor relation asked here. The present theorem is independent of their machinery and works for every prime p .

6 Verification and novelty

The proof was checked at the three points where information could be lost: the inclusion–exclusion reconstruction of the whole Boolean array law, the period/kernel equality for surjective maps, and the isolation of the affine orbit from the full-rank stratum. No computation is used as proof.

The four lightweight run indexes had no entry for arXiv:1308.4108 or this finite-cylinder classification. Bounded primary-arXiv searches on 13 August 2026 used the exact title, exact uniqueness wording, finite-field function limits, nilspace graphons, and affine exchangeability. They found the source, the 2014 thesis version, and arXiv:2203.08915, but no direct finite-core theorem or full answer in the source parametrization. Novelty confidence is moderate.

Human-review recommendation. Review as a clean substantial partial theorem. The main checks are the identification of consistent degree-one samples with uniform affine maps and the claim that equality of all $t_{\mathcal{L}}$ values supplies every subset density needed by inclusion–exclusion.

References

- [1] H. Hatami, P. Hatami, and J. Hirst, *Limits of Boolean Functions on $\mathbb{F}_p^n$* , arXiv:1308.4108 (2013).
- [2] P. Candela, D. González-Sánchez, and B. Szegedy, *On $\mathbb{F}_2^\omega$ -affine-exchangeable probability measures*, arXiv:2203.08915 (2022).